Day 01

Updated Content: Number System



Campusmonk

Placement & Competitive Exams(IBPS – IT Officer, Other Exams)

Concept Clarity Sheet

Question 1:

If $\frac{3}{4}$ of the difference of $2\frac{1}{4}$ and $1\frac{2}{3}$ is subtracted from $\frac{2}{3}$ of $3\frac{1}{4}$ the result is

A. $\frac{-48}{83}$

B. $\frac{48}{83}$

C. $\frac{-83}{48}$

D. $\frac{83}{48}$

Question 2 :

P is the product of all prime numbers from 1 to 100. Then the number of zeros at the end of the product is

(a)0

(b)1

(c)24

(d) None of these

Concept Clarity Sheet

Question 3:

What is the higher power of 2 in 1! + 2! + 3! ..... 100!?

24 (b) 3 (c) 0 (d) 97

Question 4:

If n= 1+x, where x is the product of four consecutive positive integers, then which of the following statement is/are true

1. n is odd

2. n is prime

3.n is a perfect square

A. 1 only
B. 2 only
C. 3 only
D. 1&3 only

Question 5:

What is the maximum value of m if the number

$$N = 35x45x55x60x124x75$$

is divisible by 5^m ?

(a)4 (b) 5 (c) 6 (d) 7

Question 06:

Let a, b, c be distinct digits. Consider a two digit number 'ab' and a three digit number 'ccb'.

If (ab)αb^2 = ccb and ccb > 300

then the value of b is CAT 2003

A. 0

B. 1

C. 5

D. 6

E. 7

Question 07:

The product of two consecutive even number is 9408 which of the following is the greatest number?

A. 96

B. 94

C. 92

D. 90

E. None of these

Question 08:

The sum of the digits of a two-digit number is 15 and the difference between the digits is 3. What is the two-digit number?

A.69

B.78

C.96

D. Cannot be determined

E. None of these

Question 09:

In a two-digit number, the digit at the unit's place is 1 less than twice the digit at the ten's place. If the digits at unit's and ten's place are interchanged, the difference between the new and the original number is less than the original number by 20. The original number is

(a) 59

(b) 23

(c) 35

(d) 47

Question 10:

In a two digit positive number, the digit in the units place is equal to the square of the digit in tens place, and the difference between the number and the number obtained by interchanging the digits is 54. What is 40% of the original number?

A. 15.6
B. 39
C. 37.2
D. 24

Question 11:

Find the maximum number of trees which can be planted 10m apart on two sides of a straight road 1860m long?

TITA

Question 12:

A man engaged a servant on the condition that he would pay him Rs. 90 and a turban after a service of one year. He served only for nine months and received the turban and Rs. 65. The price of the turban is:

TITA

Question 13:

In a division sum, the divisor is 10 times the quotient and 5 times the remainder. If the remainder is 48, then the dividend is:

1. 5808
2. 5809
3. 5484
4. 3472

Question 14:

Consider a sequence, where the nth term is $t_n = \frac{n}{(n+2)}$, n=1, 2..

The value of $t_3 * t_4 * t_5 * \ldots t_{53}$ equals

a. $\frac{2}{495}$ b. $\frac{2}{477}$ c. $\frac{12}{55}$ d. $\frac{1}{1485}$ e. $\frac{1}{2970}$

Question 15:

If x and y are positive integers, is y odd?

Statement 1: x is odd.

Statement 2: xy is odd.

A. Statement 1 alone is sufficient
B. Statement 2 alone is sufficient
C. Both statement 1 and statement 2 together are sufficient
D. Both statement 1 and statement 2 even together are not sufficient

Question 16:

Is ba^b + b^{b+1} is even ? (If a , b are positive integers)

I. If b is even

II. If a is even

A. Statement I alone is sufficient
B. Statement II alone is sufficient
C. Both statement I and statement II together are sufficient
D. Both statement I and statement II even together are not sufficient
E. Either I alone or II alone is sufficient to answer the question .

Question 17:

How many keystrokes are needed to type numbers from 1 to 1000?

A. 2704

B. 2890

C. 2893

D. 3001

E. None of these

Question 18:

1. Which of the following fraction is the smallest?

$$\frac{7}{6}, \frac{7}{9}, \frac{4}{5}, \frac{5}{7}$$

(1) $\frac{7}{6}$ (2) $\frac{7}{9}$

(3) $\frac{4}{5}$ (4) $\frac{5}{7}$

Question 19:

5. Which of the following is the largest fraction ? $\frac{6}{7}, \frac{5}{6}, \frac{7}{8}, \frac{4}{5}$

(1) $\frac{6}{7}$ (2) $\frac{4}{5}$

(3) $\frac{5}{6}$ (4) $\frac{7}{8}$

Question 20:

. Which of the following is correct ?

(1) $\frac{2}{3} < \frac{3}{5} < \frac{11}{15}$

(2) $\frac{3}{5} < \frac{2}{3} < \frac{11}{15}$

(3) $\frac{11}{15} < \frac{3}{5} < \frac{2}{3}$

(4) $\frac{3}{5} < \frac{11}{15} < \frac{2}{3}$

Question 21:

Arrange $\frac{4}{5}, \frac{7}{8}, \frac{6}{7}, \frac{5}{6}$ in the ascending order :

(1) $\frac{4}{5}, \frac{7}{8}, \frac{6}{7}, \frac{5}{6}$ (2) $\frac{5}{6}, \frac{6}{7}, \frac{7}{8}, \frac{4}{5}$

(3) $\frac{4}{5}, \frac{5}{6}, \frac{6}{7}, \frac{7}{8}$ (4) $\frac{7}{8}, \frac{6}{7}, \frac{5}{6}, \frac{4}{5}$

Question 22:

Sum of two numbers is 40 and their product is 375. What will be the sum of their reciprocals?

(1) $\frac{8}{75}$

(2) $\frac{1}{40}$

(3) $\frac{75}{8}$

(4) $\frac{75}{4}$